

Balanceè SmartPump

Everything the SmartPump V1 team needs — tracker, docs, and execution plan — in one place.

≡ Show menu

BALANCEÈ SMARTPUMP · ANDROID SCREEN SPEC

Every screen. *Every state. Every flow.*

Complete visual specification for Ekomobong. All 5 transaction flows, all pump display states, all attendant UI actions. Build exactly what you see here.

Idle + Mode Select

Flow 1 — Fixed Pre-pay

Flow 2 — Fill-up Cash

Flow 3 — Fill-up Digi

FLOW 5 · USSD — ZERO INTERNET

No data. No bank app. *Just a ss-phone.*

Customer dials a USSD code. Bank processes it and sends an SMS to the pump unit's SIM. Android parses the SMS and unlocks the relay. Works on any 2G ss-phone. No internet on either device required.

SCREEN SEQUENCE — USSD OFFLINE FLOW

USSD CODE DISPLAYED

WAITING FOR SMS

SMS RECEIVED +

USSD OFFLINE 0

DIAL THIS CODE

***737*5000*847#**

GTBank · ₦5,000 · Ref: 847

OTHER BANKS

Access	*901*5000*847#
Zenith	*966*5000*847#
UBA	*919*5000*847#

Pump shows USSD codes for 3 major banks. Customer dials on

AWAITING SMS

WAITING FOR SMS CONFIRMATION

Bank sends SMS after USSD completes. Usually 10-30 seconds.

AMOUNT	₦5,000
REF	847
SIM STATUS	MTN · Signal

SMS CONFIRMED

LITRES DIS

2.1

of 5.75L au

METHOD

OFFLINE

SMS parsed. Re

Counting n

their ss-phone.

Android monitors pump unit SIM
for incoming SMS containing the
reference.

SMS parsing — what to look for

The Android unit must listen for incoming SMS on the pump SIM. Each bank sends a different SMS format. The parser must extract: (a) confirmation that payment succeeded, (b) the amount, (c) the transaction reference that matches the pending transaction.

```
// GTBank example SMS:  
// "Your GTBank acct was debited ₦5,000.00 for BALANCEE-847 on 30/04/26..."  
// Extract: amount=5000, ref=847  
  
// Android SMS listener:  
SmsReceiver extends BroadcastReceiver {  
    onReceive(context, intent) {  
        String sms = intent.getStringExtra("pdu_parsed");  
        PaymentConfirmation p = SMSParser.parse(sms);  
        if (p.matches(pendingTxn)) authorisePump(p.amount);  
    }  
}
```

SMS patterns differ per bank. Start with GTBank (*737#) for V1. Add Access and Zenith before field test. Full Nigerian bank coverage in V2.

V1 REQUIRED — offline coverage for low-connectivity stations

